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S-MPBART: Soft Multinomial Probit Bayesian Additive Regression Trees

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Abstract

We propose soft multinomial probit Bayesian additive regression trees (S-MPBART) as a new multiclass classification method. This method builds on the multinomial probit extension of Bayesian additive regression trees, MPBART, by integrating soft decision trees and a sparsity-inducing prior in its framework. These components enable S-MPBART to adapt to smooth latent structures in the multinomial probit model while simultaneously performing variable selection. Empirical results from four simulation studies show that S-MPBART performs very well in the case of smooth latent structures, while remaining competitive with existing methods when such smoothness is absent. Moreover, the results highlight the effectiveness of the sparsity-inducing prior in high-dimensional settings with many irrelevant predictors. Across a range of real-data examples, S-MPBART is competitive with standard tree classifiers, and consistently improves on MPBART. These results demonstrate that S-MPBART is a flexible and widely applicable Bayesian tree-based approach for multiclass classification, with practical advantages in settings involving smooth and nonlinear latent structures or high-dimensional predictor spaces.

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Chapter 1

Introduction

Multiclass classification is a fundamental task in supervised machine learning, where the goal is to assign observations to one of $K > 2$ discrete classes. Formally, given a set of training indices $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where each $\mathbf{x}_i \in \mathcal{X} \subseteq \mathbb{R}^p$ denotes a p-dimensional predictor vector and $y_i \in \mathcal{Y} = \{1, \ldots, K\}$ is the corresponding class label, the goal is to learn a function $f : \mathcal{X} \to \mathcal{Y}$ that generalizes well to unseen data. Many real-world problems involve more than two outcome classes, with example applications in disease diagnosis (Detrano et al., 1989; Sajda, 2006), image and speech recognition (Yu & Deng, 2016; Pak & Kim, 2017; Shen et al., 2017), sports analytics (Bunker & Thabtah, 2019), sentiment analysis (Bouazizi & Ohtsuki, 2019) and marketing (van Wezel & Potharst, 2007; Kansal et al., 2018).

Among foundational linear approaches for multiclass classification are linear discriminant analysis (Fisher, 1936), the multinomial logit model (McFadden, 1973) and the multinomial probit model (McFadden, 1989). The latter two are rooted in discrete choice theory, with the multinomial probit model allowing for correlated alternatives. These models are valued for their simplicity, interpretability, and strong theoretical foundations. However, they rely on the assumption of a linear relationship between the predictors and the outcome probabilities, which limits their ability to capture complex nonlinear patterns. Moreover, they do not inherently model interactions between predictors, and therefore require manual feature engineering to represent such effects. These limitations highlight the need for more flexible, nonlinear approaches that can automatically learn complex relationships and interactions commonly present in practical classification problems.

One class of models that addresses these limitations is tree-based methods, which offer a flexible approach to modeling nonlinearities and interactions without requiring a parametric model. A foundational method in this class is the Classification and Regression Trees (CART) algorithm (Breiman et al., 1984; Quinlan, 1986), which recursively splits the predictor space to form a single decision tree. Building on this idea, ensemble methods like Random Forest (Breiman, 2001) and Gradient Boosted Trees (Friedman, 2001) aggregate multiple trees to improve predictive performance and stability. These methods have been adopted widely because of their accuracy and robustness in a wide range of applications. An alternative line of research introduces Bayesian approaches to tree-based modeling. As the Bayesian counterpart of the CART algorithm, Bayesian CART (Chipman et al., 1998) places a prior distribution over tree components, which is updated by the data to obtain a posterior distribution over all possible trees. Extending the single-tree framework, Bayesian Additive Regression Trees (BART) (Chipman et al., 2010) is an ensemble of Bayesian trees which models the response as a sum of many trees, combined with a regularization prior shrinking the influence of individual trees. Empirically, BART has demonstrated strong predictive performance for both regression and classification tasks across a wide range of applications (Chipman et al., 2010; Zhang & Härdle, 2010; Bonato et al., 2011).

Since its introduction, extensions of BART have been developed for many different econometric and statistical models (e.g. Chakraborty, 2016; Huber & Rossini, 2022; O'Neill, 2024), where the sum-of-trees framework is used to replace usual parametric structures. In this paper we turn our focus to the SoftBART (SBART) framework (Linero & Yang, 2018). While the aforementioned extensions of BART represent applications of BART to specific econometric frameworks, they do not alter the internal mechanics of the BART model itself. In contrast, SBART constitutes a methodological change of the core BART framework by replacing standard decision trees with soft decision trees. Unlike traditional decision trees, which partition the predictor space using hard, axis-aligned splits, soft decision trees introduce smooth probabilistic splits at each node. Instead of sending each observation deterministically left or right down the tree based on a hard splitting rule, soft decision trees assign probabilities to both directions. In this way, each leaf node is associated with a probability that reflects the likelihood of an observation ending up in that leaf node, which is determined by the product of probabilistic splitting decisions at each internal node along the path to the leaf. The decision trees in SBART are conceptually aligned to the work of Irsoy et al. (2012), which considers a soft variant of the CART algorithm. However, the two approaches differ in multiple aspects, most notably in how the trees are learned and in the model structure. While Irsoy et al. (2012) employ a greedy optimization procedure to learn a single soft decision tree, SBART builds on the Bayesian backfitting strategy of Chipman et al. (2010) to learn an ensemble of such trees. In addition to the use of soft decision trees, the SBART model incorporates a sparsity-inducing prior (Linero, 2018) to overcome the curse of dimensionality, a challenge commonly faced by tree-based models.

While SBART has shown good empirical performance (Linero & Yang, 2018), it has so far been developed primarily for regression tasks. To date, the framework has not been extended to multiclass classification settings. To address this gap, we build on the multinomial probit BART (MPBART) framework (Kindo et al., 2016; Xu et al., 2024), which combines the multinomial probit model with BART to handle multiclass outcomes. MPBART models the latent utilities associated with outcome classes as sum-of-trees structures, and extends the BART prior to be consistent with the higher-dimensional setting. However, MPBART relies on hard decision trees as in standard BART. In this paper, we introduce soft MPBART (S-MPBART) as a new multiclass classification method which incorporates both soft decision trees and a sparsity-inducing prior into the MPBART framework. Based on a comprehensive simulation study with a variety of data-generating schemes, we find that S-MPBART attains improved predictive performance compared to existing methods when the underlying latent structures behave smoothly, while remaining competitive even when such smoothness is absent. Furthermore, we find that S-MPBART is capable of performing effective variable selection through its sparsity-inducing prior. When applied to real datasets, S-MPBART consistently improves on MPBART, and proves to be a strong competitor to other methods.

The paper proceeds as follows. Section 2 provides a brief review of the standard BART model and introduces the proposed S-MPBART methodology. Section 3 and Section 4 evaluate the performance of S-MPBART through a simulation study and applications to real datasets, respectively. Section 5 discusses the main findings and concludes the paper.

Chapter 2

Methodology

2.1 Review of BART

2.1.1 Model and prior specification

Consider the fundamental problem of approximating an unknown function f, which predicts a continuous outcome y using a p-dimensional vector of covariates $\mathbf{x} = (x_1, \ldots, x_p)$ as

$$y_i = f(\mathbf{x}_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2), \tag{2.1}$$

for observations $i = 1, \ldots, n$. Chipman et al. (2010) introduce the idea to approximate $f(\mathbf{x})$ by a sum-of-trees model

$$f(\mathbf{x}) \approx \sum_{j=1}^{m} g(\mathbf{x}; T_j, M_j), \tag{2.2}$$

where T_j denotes a binary tree associated with a set of decision rules and $M_j = \{\mu_{jl} : l = 1, \ldots, L_j\}$ denotes the set of leaf node parameters corresponding to a tree T_j with L_j leaf nodes. The decision rules are of the form $\{x_h \leq C\}$ versus $\{x_h > C\}$, where x_h is a single component of $\mathbf{x}$ and C is a value which falls in the range of x_h. An input $\mathbf{x}$ is associated with a single leaf node by following a path down the tree induced by a sequence of such decision rules. The function $g(\mathbf{x}; T_j, M_j)$ assigns the corresponding leaf node parameter to $\mathbf{x}$. As a consequence, each tree contributes to the final prediction in an additive way. The representation in Equation 2.2 allows for flexible modeling of complex and highly nonlinear relationships between the outcome and the covariates. Furthermore, it allows for the inclusion of interaction effects between different covariates. Although the model is typically not identified as different sets of trees can result in the same regression function f, this is not a major concern as individual trees are usually not of interest.

Chipman et al. (2010) assume prior independence between trees T_j and between leaf node parameters μ_{jl}. The prior is specified as follows:

$$p(T_1, \ldots, T_m, M_1, \ldots, M_m, \sigma) \propto \left[\prod_{j=1}^{m} p(T_j) \left[\prod_{l=1}^{L_j} p(\mu_{jl} | T_j) \right] \right] p(\sigma). \tag{2.3}$$

The tree prior $p(T_j)$ is specified in the following way. The probability of a node η being nonterminal equals $\gamma(1+d_\eta)^{-\beta}$, where d_η denotes the depth of node η. Here $\gamma \in (0,1)$ and $\beta \in [0,\infty)$ govern the size and shape of tree T_j, respectively. This generates a branching process which terminates almost surely for positive β (Athreya & Ney, 2004), and favors trees for which terminal nodes do not vary too much in depth (Chipman et al., 1998). At each internal node, Chipman et al. (2010) choose both the splitting variable and splitting value uniformly over the set of available variables and values, that is, splitting variables and values which do not lead to empty leaf nodes.

The leaf node parameters are given a conjugate normal prior $\mu_{jl} \sim N(\mu_\mu, \sigma_\mu^2)$. By scaling the observed outcome y to fall in the interval $[-0.5, 0.5]$, the prior on μ_{jl} can be centered around $\mu_\mu = 0$ such that $\mu_{jl} \sim N(0, \sigma_\mu^2)$, where $\sigma_\mu = 0.5/(e\sqrt{m})$. Here e is a suitable positive constant, usually set at $e = 2$. This prior has the effect of shrinking the leaf node parameters towards zero, regularizing the influence of individual trees in the sum-of-trees structure.

Finally, Chipman et al. (2010) use a conjugate inverse chi-squared prior for the error variance $\sigma^2 \sim \nu\lambda/\chi^2_\nu$, where the hyperparameters ν and λ are calibrated using a data-informed approach.

2.1.2 Posterior computation

To obtain samples from the posterior $p((T_1, M_1), \ldots, (T_m, M_m), \sigma|\mathbf{y})$, Chipman et al. (2010) propose a Metropolis-within-Gibbs sampler, in the form of a backfitting MCMC algorithm (Hastie & Tibshirani, 2000).

Denote $\mathcal{T}_{(-j)}$ as the set of all trees except tree T_j, and similarly denote $\mathcal{M}_{(-j)}$ as the corresponding leaf node parameters. The sampler involves m successive draws of $(T_j, M_j)|\mathcal{T}_{(-j)}, \mathcal{M}_{(-j)}, \sigma, \mathbf{y}$ for $j = 1, \ldots, m$ followed by a draw from $\sigma|T_1, \ldots, T_m, M_1, \ldots, M_m, \mathbf{y}$, the full conditional of σ. The latter draw is a standard inverse gamma distribution draw. Regarding the tree and terminal node parameter draws, it can be shown that T_j and M_j only depend on $(\mathcal{T}_{(-j)}, \mathcal{M}_{(-j)}, \mathbf{y})$ through partial residuals $\mathbf{r}_j := \mathbf{y} - \sum_{k \neq j} g(\mathbf{x}; T_k, M_k)$. Therefore each of the m successive draws can be split into two separate draws:

$$T_j|\mathbf{r}_j, \sigma, \tag{2.4}$$

$$M_j|T_j, \mathbf{r}_j, \sigma. \tag{2.5}$$

The tree draw in Equation 2.4 is obtained by the Metropolis-Hastings procedure described in Chipman et al. (1998). Finally, the M_j draw in Equation 2.5 entails a set of independent draws of leaf node parameters μ_{jl}, $l = 1, \ldots, L_j$, from a normal distribution.

After a suitable burn-in period, Q iterations of the backfitting MCMC algorithm generate a sequence $f_1^*, \ldots, f_Q^*$, where $f_q^*(\cdot) = \sum_{j=1}^m g(\cdot; T_j^*, M_j^*)$ for $q = 1, \ldots, Q$. The resulting sequence may be regarded as an approximate, dependent sample of size Q from the posterior distribution $p(f|\mathbf{y})$. This sample can subsequently be used to compute posterior inference statistics, including the posterior mean and credible intervals. For further details on the BART model specification and implementation of the backfitting MCMC algorithm, see Chipman et al. (2010), Kapelner & Bleich (2016) and Tan & Roy (2019).

2.2 Soft MPBART

2.2.1 Model specification

We first introduce the notation of the multinomial probit (MNP) model. Consider a multinomial outcome y_i that takes values in $\{0, \dots, K\}$ for observations $i = 1, \dots, n$. For example, y_i may represent the choice of an economic agent between different choice options, or denote the selected category among a set of mutually exclusive classes. We are interested in estimating the conditional probability $p(y_i = k|\mathbf{x}_i)$ for $k = 0, \dots, K$. The observed outcome y_i arises from a set of underlying latent utilities $\mathbf{w}_i = (w_{i0}, \dots, w_{iK}) \in \mathbb{R}^{K+1}$, such that $y_i = \arg\max_k w_{ik}$. In this form, y_i is invariant to both translation and scaling of $\mathbf{w}$ by a positive constant. To identify the location of the latent utility system we can model y_i in terms of latent variables $\mathbf{z}_i = (z_{i1}, \dots, z_{iK}) \in \mathbb{R}^K$ (e.g. Albert & Chib, 1993; McCulloch et al., 2000; Imai & van Dyk, 2005) where $z_{ik} = w_{ik} - w_{i0}$ for $k = 1, \dots, K$. This implies

$$y_i = \begin{cases} 0, & \text{if } \max(\mathbf{z}_i) < 0, \\ k, & \text{if } \max(\mathbf{z}_i) = z_{ik}, \end{cases} \tag{2.6}$$

where $\max(\mathbf{z}_i)$ is the maximum element of $\mathbf{z}_i = (z_{i1}, \dots, z_{iK})$. Note that in the definition of the z_{ik} variables and in Equation 2.6, we assume without loss of generality that category 0 acts as the reference alternative.

The latent variables depend on covariates $\mathbf{x}_i = (x_{i1}, \dots, x_{ip}) \in \mathbb{R}^p$ as follows:

$$\mathbf{z}_i = \mathbf{G}(\mathbf{x}_i; \boldsymbol{\theta}) + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim MVN(\mathbf{0}, \boldsymbol{\Sigma}) \quad \text{for } i = 1, \dots, n, \tag{2.7}$$

where $\mathbf{G}(\mathbf{x}_i; \boldsymbol{\theta}) = (G_1(\mathbf{x}_i, \boldsymbol{\theta}_1), \dots, G_K(\mathbf{x}_i, \boldsymbol{\theta}_K))$ is a vector of K functions, $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K)$ denote the corresponding parameters and $\boldsymbol{\Sigma}$ is a $K \times K$ covariance matrix. With this specification the latent variables are effectively modeled by a multivariate normal distribution, which allows for correlation across different outcome levels. The classical MNP framework assumes a linear model for each component of the latent variable, that is $G_k(\mathbf{x}_i, \boldsymbol{\theta}_k) = \mathbf{x}_i'\boldsymbol{\theta}_k$ for $k = 1, \dots, K$. Although this specification provides good interpretability and facilitates easy computation, it is unable to capture more complex nonlinear and interaction effects between the outcome and the covariates. Kindo et al. (2016) proposed multinomial probit BART (MPBART) as a solution to this problem. The idea is to specify each G_k function as a sum-of-trees function, such that the mean of each component of the latent vector $\mathbf{z}_i$ is approximated by a sum of m regression trees. This model serves as the foundation to build our S-MPBART methodology on, and it reads as follows:

$$\mathbf{z}_i = \mathbf{G}(\mathbf{x}_i; \boldsymbol{\mathcal{T}}, \boldsymbol{\mathcal{M}}) + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim MVN(\mathbf{0}, \boldsymbol{\Sigma}) \quad \text{for } i = 1, \dots, n, \tag{2.8}$$

where $\mathbf{G}(\mathbf{x}_i; \boldsymbol{\mathcal{T}}, \boldsymbol{\mathcal{M}}) = (G_1(\mathbf{x}_i; \mathcal{T}_1, \mathcal{M}_1), \dots, G_K(\mathbf{x}_i; \mathcal{T}_K, \mathcal{M}_K))$ is a vector of K sum-of-trees functions. Here $\mathcal{T}_k = \{T_{kj} : j = 1, \dots, m\}$ denotes the set of trees used in the sum-of-trees model for the k'th component of $\mathbf{z}_i$. The corresponding sets of leaf node parameters are denoted by $\mathcal{M}_k = \{M_{kj} : j = 1, \dots, m\}$. As in Chipman et al. (2010), the sum-of-trees functions are specified as

$$G_k(\mathbf{x}_i; \mathcal{T}_k, \mathcal{M}_k) = \sum_{j=1}^{m} g(\mathbf{x}_i; T_{kj}, M_{kj}) \quad \text{for } k = 1, \ldots, K, \tag{2.9}$$

where T_{kj} denotes the j'th tree in the k'th sum of trees, $M_{kj} = \{\mu_{kjl} : l = 1, \ldots, L_{kj}\}$ denotes the corresponding leaf node parameters, and L_{kj} denotes the number of leaf nodes in tree T_{kj}. Combining Equations 2.8 and 2.9 yields the following specification:

$$\begin{pmatrix} z_{i1} \\ z_{i2} \\ \vdots \\ z_{iK} \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^{m} g(\mathbf{x}_i; T_{1j}, M_{1j}) \\ \sum_{j=1}^{m} g(\mathbf{x}_i; T_{2j}, M_{2j}) \\ \vdots \\ \sum_{j=1}^{m} g(\mathbf{x}_i; T_{Kj}, M_{Kj}) \end{pmatrix} + \begin{pmatrix} \varepsilon_{i1} \\ \varepsilon_{i2} \\ \vdots \\ \varepsilon_{iK} \end{pmatrix}, \quad \boldsymbol{\varepsilon}_i \sim N(0, \boldsymbol{\Sigma}), \tag{2.10}$$

which is the MPBART model as in Kindo et al. (2016). Note that each component of the latent vector $\mathbf{z}_i$ has a separate sum-of-trees model, but due to the error structure they can be correlated.

In most tree-based methods, including MPBART, a tree is associated with a set of hard splitting rules. Such splitting rules are of the form $\{x_h \leq C_\eta\}$ versus $\{x_h > C_\eta\}$ where x_h is a single component of the input $\mathbf{x}$ and C_η is the split value corresponding to internal node η, which lies within the range of x_h. The predictor x_h is selected with probability $s_{kh} \in \mathbf{s}_k = (s_{k1}, \ldots, s_{kp})$, where $\mathbf{s}_k$ denotes the vector of splitting probabilities for sum-of-trees model $k = 1, \ldots, K$. In this way, each $\mathbf{x}$ is assigned a deterministic path down the tree, ending up in a single leaf node which represents a unique and nonoverlapping partition of the covariate space. For our S-MPBART method, we diverge from this hard splitting framework and instead focus on soft splits, which were originally introduced in a BART framework by Linero & Yang (2018). In a MNP context, this allows for the covariates to have smoother effects on the outcome probabilities. Contrary to the standard BART framework, where an input $\mathbf{x}$ follows a deterministic path to a single leaf node, the soft framework replaces this hard assignment with a probabilistic one. Instead of assigning $\mathbf{x}$ to a single partition, each leaf node receives a weight representing the probability that $\mathbf{x}$ belongs to that partition of the covariate space. The prediction for the outcome y is then computed as a weighted average of the leaf node parameters, using these probabilistic weights. Following Linero & Yang (2018), we first specify the probability of $\mathbf{x}$ going left down the tree at an internal node η which belongs to a tree $T \in \mathcal{T}_k$ as

$$\psi(\mathbf{x}; \mathcal{T}_k, \eta) = \psi\left(\frac{x_h - C_\eta}{\tau_{k,\eta}}\right), \quad k = 1, \ldots, K. \tag{2.11}$$

Note that in the soft tree framework $\mathbf{x}$ does not actually follow a path through the tree. However, an expression as in Equation 2.11 is necessary to compute the leaf weights. The function $\psi(\cdot)$ is a gating function which we specify as the logistic function and $\tau_{k,\eta}$ is a bandwidth parameter which controls the sharpness of the decision. As $\tau_{k,\eta} \to 0$, the model approaches a hard decision, whereas the decision becomes completely random as $\tau_{k,\eta} \to \infty$. The weight allocated to leaf l

belonging to a tree $T \in \mathcal{T}_k$ then equals

$$\phi(\mathbf{x}; \mathcal{T}_k, l) = \prod_{\eta \in A(l)} \psi(\mathbf{x}; \mathcal{T}_k, \eta)^{1-V_\eta}(1 - \psi(\mathbf{x}; \mathcal{T}_k, \eta))^{V_\eta}, \quad (2.12)$$

where $A(l)$ denotes the set of ancestor nodes of leaf l and V_η indicates whether the path to l goes right ($V_\eta = 1$) or left ($V_\eta = 0$) at η. Although $\mathbf{x}$ does not actually follow a path down the tree, Equation 2.12 can be interpreted as the probability that $\mathbf{x}$ ends up at leaf l. Whereas in hard trees each leaf node only contributes locally within its corresponding subregion of the covariate space, the probabilistic nature of soft trees allows each leaf node to contribute to the prediction across the entire input space. This enables soft trees to use information across different covariate regions, with the degree of smoothing controlled by the bandwidth parameters $\tau_{k,\eta}$.

2.2.2 Prior specification

The complete set of parameters of S-MPBART is given by

$$\mathbf{\Theta} = \{(T_{kj}, M_{kj})_{k=1,\dots,K,j=1,\dots,m}, \mathbf{\Sigma}, (\mathbf{s}_k)_{k=1,\dots,K}, (\tau_{kj})_{k=1,\dots,K,j=1,\dots,m}, (\sigma_k, a_k)_{k=1,\dots,K}\}, \quad (2.13)$$

where σ_k and a_k are parameters used in the priors of M_{kj} and $\mathbf{s}_k$, respectively. We now provide detailed prior specifications for all components of $\mathbf{\Theta}$.

The T_{kj} and M_{kj} priors. Following Chipman et al. (2010), we specify a branching process prior for each tree T_{kj}, with the probability that a node η at depth d_η is nonterminal equal to $\gamma(1 + d_\eta)^{-\beta}$, and use the recommended default values $\gamma = 0.95$ and $\beta = 2$. Kindo et al. (2016) extend the prior on leaf node parameters of Chipman et al. (2010) to a multivariate setting, such that

$$\mu_{kjl}|T_{kj} \sim N(\mu_k, \sigma_k^2) \quad \text{for } k = 1, \dots, K. \quad (2.14)$$

Regarding the parameters μ_k and σ_k, we set $\mu_k = 0$ and specify a hyperprior on σ_k. Instead of fixing σ_k like in Kindo et al. (2016), for each k, we follow Linero & Yang (2018) by giving σ_k a half-Cauchy prior, $\sigma_k \sim \text{Cauchy}_+(0, 0.25)$. This approach for σ_k is more suitable in the context of soft trees, since fixing σ_k would impose uniform shrinkage across all leaf nodes, disregarding the fact that in soft trees, the uncertainty and influence associated with each leaf node can vary substantially due to the probabilistic nature of the trees. As an initial guess, we follow the BART prior by setting $\sigma_k = 0.5/(e\sqrt{m})$ for all k, and use $e = 1/3$ to obtain a plausible value for the multinomial probit context (Kindo et al., 2016).

The $\mathbf{\Sigma}$ prior. The model in Equation 2.10 suffers from a well-known identifiability issue, as multiplying both sides by a positive scalar does not affect the resulting choice outcome (Keane, 1992; McCulloch & Rossi, 1994; McCulloch et al., 2000). Multiple solutions have been proposed to make the model identifiable. Among others, McCulloch & Rossi (1994) and Imai & van Dyk (2005) restrict the first diagonal element of $\mathbf{\Sigma}$ to one. Burgette & Nordheim (2012) propose a different restriction, namely constraining the trace of $\mathbf{\Sigma}$ to equal K. Like Kindo et al. (2016), we opt to implement the trace restriction. This avoids introducing asymmetry and leads to better mixing and convergence properties of the MCMC algorithm (Imai & van Dyk, 2005). Note that with the trace restriction, it is not possible to define a conjugate prior for $\mathbf{\Sigma}$. Instead, we define a conjugate inverse-Wishart prior for an unconstrained covariance matrix $\mathbf{\Sigma}^*$, that is, $\mathbf{\Sigma}^* \sim IW(\nu_0, \mathbf{S}_0)$. We set the prior degrees of freedom ν_0 to $K+2$ and the prior scale matrix as $\mathbf{S}_0 = I_K$, where I_K denotes the $K \times K$ identity matrix. Samples of $\mathbf{\Sigma}$ can then be obtained by rescaling samples of the full conditional of $\mathbf{\Sigma}^*$ in such a way that the trace restriction is satisfied.

The $\mathbf{s}_k$ *and* τ_{kj} *priors.* We extend the priors used in Linero & Yang (2018) to a multivariate setting. For each of the splitting probability vectors $\mathbf{s}_k$, we specify a sparsity-inducing Dirichlet prior (Linero, 2018):

$$\mathbf{s}_k \sim \mathrm{Dir}(a_k/p, \ldots, a_k/p) \quad \text{for } k = 1, \ldots, K, \tag{2.15}$$

where the parameter a_k represents the expected amount of sparsity. Following Linero & Yang (2018), we specify a hyperprior $a_k/(a_k + p) \sim \mathrm{Beta}(0.5, 1)$, which provides a good prior balance between sparse and non-sparse settings. Note that the parameters a_k can be different between the sum-of-trees models. Regarding the bandwidth parameters τ_{kj}, we assume that within each single tree T_{kj}, τ_{kj} is shared between all internal nodes, but across trees τ_{kj} may differ, leading to τ being indexed by both $k = 1, \ldots, K$ and $j = 1, \ldots, m$. The prior is specified as

$$\tau_{kj} \sim \mathrm{Exp}(0.1) \quad \text{for } k = 1, \ldots, K \text{ and } j = 1, \ldots, m, \tag{2.16}$$

which allows for a wide variety of possible shapes for the gating function $\psi(\cdot)$.

The last remaining specification is the number of trees m. We assume that each sum-of-trees model in Equation 2.10 has the same number of trees in the ensemble. Although various methods for tuning m exist (e.g. Watanabe, 2013; Vehtari et al., 2017), having the number of trees fixed to a default value of $m = 200$ has been shown to achieve good performance on most data sets (Chipman et al., 2010; Kindo et al., 2016; Linero & Yang, 2018).

Finally, following Linero & Yang (2018), we preprocess $\mathbf{x}_i$ by applying a rank normalization. Each x_{ih} is mapped to its rank, with $\min_i x_{ih} = 1$ and $\max_i x_{ih} = n$, followed by a linear transformation such that the transformed covariates are in the interval $[0, 1]$. This preprocessing ensures that the prior remains invariant to monotone transformations of the covariates, and is necessary for maintaining numerical stability in the soft decision functions, where the logistic gating mechanism is sensitive to the scale of the inputs.

2.2.3 Posterior computation and inference

In Algorithm 1 we describe a high-level posterior sampling scheme for S-MPBART. The initializations and run settings of S-MPBART are provided in Appendix A.

The first step is sampling the latent vector $\mathbf{z}_i$ from a truncated multivariate normal distribution. Given the observed outcome y_i, the definition in Equation 2.6 implies a set of linear constraints on $\mathbf{z}_i$ which defines the polyhedral truncation region of the sampling distribution. Specifically, for each dimension k the value $\max\{0, \max(\mathbf{z}_{i(-k)})\}$ serves as a lower truncation point if $y_i = k$, and as an upper truncation point if $y_i \neq k$.

To sample the trees and leaf node parameters (T_{kj}, M_{kj}) for $k = 1, \ldots, K$ and $j = 1, \ldots, m$, we first need to transform the sampled latent variables to remove the dependence between latent dimensions, so that each of the K sum-of-trees models in Equation 2.10 can be trained on conditionally independent targets consistent with the multinomial probit structure. For

$i = 1, \ldots, n$ and $k = 1, \ldots, K$ we obtain the transformed latent variables

$$z^*_{ik} = z_{ik} - \left[\mathbf{z}_{i(-k)} - \mathbf{G}_{(-k)}(\mathbf{x}_i; \mathcal{T}_k, \mathcal{M}_k)\right]' \boldsymbol{\Sigma}^{-1}_{(-k)(-k)} \boldsymbol{\Sigma}_{(-k)k}, \tag{2.17}$$

where $\mathbf{z}_{i(-k)}$ denotes the sampled latent variables for observation i excluding the k'th element and $\mathbf{G}_{(-k)}(\mathbf{x}_i; \mathcal{T}_k, \mathcal{M}_k)$ denotes the model predictions of the same latent variables, where again the k'th element is excluded. The submatrix of $\boldsymbol{\Sigma}$ with both the k'th row and column removed is denoted by $\boldsymbol{\Sigma}_{(-k)(-k)}$, and $\boldsymbol{\Sigma}_{(-k)k}$ is the k'th column of $\boldsymbol{\Sigma}$ where the k'th element is excluded. For the same reason as mentioned above, we also need to set the conditional variance of $z_k|\mathbf{z}_{(-k)}$ for each k. Letting ω_k^2 denote this conditional variance, we have

$$\omega_k^2 = \boldsymbol{\Sigma}_{kk} - \boldsymbol{\Sigma}_{k(-k)} \boldsymbol{\Sigma}^{-1}_{(-k)(-k)} \boldsymbol{\Sigma}_{(-k)k} \quad \text{for } k = 1, \ldots, K, \tag{2.18}$$

where $\boldsymbol{\Sigma}_{kk}$ is the k'th diagonal element of $\boldsymbol{\Sigma}$. With the transformed latent variables z^*_{ik} from Equation (2.17) and the corresponding conditional variances ω_k^2 from Equation (2.18), sampling of the trees and leaf node parameters for each sum-of-trees model proceeds independently for each dimension k as in standard BART, treating z^*_{ik} as the response variable and with error variance ω_k^2.

In the soft sum-of-trees models, the bandwidth parameters τ_{kj} are sampled via a Metropolis-Hastings step as described in the online supplementary materials of Linero & Yang (2018). The splitting probabilities are sampled from the full conditional $\mathbf{s}_k \sim \text{Dir}\left(a_k/p + c_{k1}, \ldots, a_k/p + c_{kp}\right)$, where $c_{ke} = \#\{\eta \in \mathcal{T}_k : \text{node } \eta \text{ splits on covariate } e\}$.

Finally, a draw of $\boldsymbol{\Sigma}$ is obtained by first drawing an unconstrained covariance matrix $\boldsymbol{\Sigma}^*$ from a standard inverted-Wishart posterior

$$\boldsymbol{\Sigma}^* \sim IW\left(\nu_0 + n, \mathbf{S}_0 + \sum_{i=1}^{n} \boldsymbol{\varepsilon}_i \boldsymbol{\varepsilon}_i'\right), \tag{2.19}$$

where $\boldsymbol{\varepsilon}_i = \mathbf{z}_i - \mathbf{G}(\mathbf{x}_i; \boldsymbol{\mathcal{T}}, \boldsymbol{\mathcal{M}})$ are the observed residuals. Consequently, the draw of $\boldsymbol{\Sigma}$ is obtained by scaling the draw of $\boldsymbol{\Sigma}^*$ with a factor $K/\text{tr}(\boldsymbol{\Sigma}^*)$, which forces the trace restriction to hold.

Algorithm 1 Sampling scheme for S-MPBART

Initialize model parameters $\mathbf{\Theta}^{(0)}$.
for n_burnin + n_sim iterations **do**
  for $i = 1, \ldots, n$ **do**
    Sample latent variables $\mathbf{z}_i$ from a truncated multivariate normal distribution.
  end for
  for $k = 1, \ldots, K$ **do**
    for $j = 1, \ldots, m$ **do**
      Sample T_{kj} via Metropolis-Hastings.
      Sample τ_{kj} via Metropolis-Hastings.
      Sample M_{kj} from a normal distribution.
    end for
    Sample $\mathbf{s}_k$ from a Dirichlet distribution.
    Sample σ_k and a_k.
  end for
  Sample $\mathbf{\Sigma}$ as follows:
  - Sample unconstrained $\mathbf{\Sigma}^*$ from an inverse-Wishart distribution.
  - Scale to obtain $\mathbf{\Sigma}$.
end for

To compare the predictive performance of S-MPBART with competing benchmark methods we use both the standard misclassification rate and the multiclass Brier score. The misclassification rate on a test data set with size N is given by

$$\frac{1}{N}\sum_{i=1}^{N} I(\hat{y}_i \neq y_i), \tag{2.20}$$

where $I(\cdot)$ denotes an indicator function, $\hat{y}_i$ denotes the predicted class and y_i is the observed class. While the misclassification rate provides a straightforward measure of overall predictive performance by counting the proportion of incorrect class labels, it does not account for the confidence or probability distribution of the predicted classes. To address this limitation, we use the multiclass Brier score as a more refined metric to evaluate predicted class probabilities, which is defined as

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{k=0}^{K}(p_{ik} - y_{ik})^2, \tag{2.21}$$

where p_{ik} is the predicted probability that $y_i = k$, and y_{ik} equals 1 if class k is the observed class for observation i, and 0 otherwise. A lower multiclass Brier score indicates better predictive performance, reflecting both higher accuracy and improved calibration of the predicted class probabilities.

Chapter 3

Simulation experiments

We perform multiple simulation experiments to test S-MPBART's performance against competing benchmark methods. The experiments explore a range of data-generating processes that vary in smoothness, the number of predictors, and the presence of noise predictors. Furthermore, the experiments are designed to illustrate the ability of S-MPBART to adapt to different levels of smoothness and to perform effective variable selection, leveraging both its soft decision trees and sparsity-inducing prior. An overview of the experiments is shown in Table 3.1.

Experiment	**# Classes**	**# Predictors**
DGP I - Local smoothness	3	2
DGP II - Global smoothness	3	10
DGP II* - Global smoothness with noise predictors	3	10 + 50 noise
DGP III - Multinomial choice	3	6

Table 3.1: Summary of the simulation experiments, reporting the number of classes and the number of predictors. *Denotes a variation of DGP II with additional noise predictors.

The benchmark methods are Random Forest (Breiman, 2001) and MPBART as implemented in Xu et al. (2024). Run settings are provided in Appendix A.

3.1 DGP I

In the first simulation setting we consider a data-generating process inspired by Park (2022), which defines two latent functions z_1 and z_2 that serve as underlying signals for a multiclass classification problem with three classes. We assume class 0 acts as the reference class. The latent variables $\mathbf{z} = (z_1, z_2)'$ are generated as

$$z_1 = \begin{cases} 5\sin(2\pi x_1) + 3x_2 + \varepsilon_1, & \text{if } x_1 \leq 0.5, \\ 5\sin(2\pi x_1) - 3x_2 + \varepsilon_1, & \text{if } x_1 > 0.5, \end{cases} \tag{3.1}$$

$$z_2 = \begin{cases} 4\cos(\pi x_2) + x_1 + \varepsilon_2, & \text{if } x_2 \leq 0.3, \\ 8\cos(\pi x_2) - x_1 - 2 + \varepsilon_2, & \text{if } x_2 > 0.3, \end{cases} \tag{3.2}$$

where $x_1, x_2 \overset{\text{i.i.d.}}{\sim} \text{UNIF}[0,1]$ and $\varepsilon_k \overset{\text{i.i.d.}}{\sim} N(0,1)$ for $k = 1, 2$. Each function is piecewise-defined with nonlinear components involving sinusoidal and cosine functions, ensuring smooth but complex relationships with two input features x_1 and x_2. The piecewise structure introduces regime-dependent behavior, creating changes in slope or intercept at threshold boundaries $x_1 = 0.5$ and $x_2 = 0.3$. These discontinuities combined with smooth trigonometric patterns contribute to nonlinearities and varying degrees of local smoothness.

The class label $y \in \{0, 1, 2\}$ follows from the generated latent variables as

$$y(\mathbf{z}) = \begin{cases} k & \text{if} \max(\mathbf{z}) = z_k > 0, \\ 0 & \text{if} \max(\mathbf{z}) < 0, \end{cases} \tag{3.3}$$

for $k = 1, 2$. This data generating process combines locally smooth nonlinear latent functions with added noise and discontinuities, creating a reasonably challenging setup for S-MPBART as it has to capture both nonlinear patterns and breaks in the underlying latent structure. In Table 3.2 we see that S-MPBART beats both benchmark methods in terms of accuracy. The benchmark methods do perform better in terms of the calibration of predictions, both showing a lower Brier score than S-MPBART, indicating slightly more confident predictions under this DGP. Nonetheless, the results demonstrate that S-MPBART remains highly competitive with both its hard-splitting counterpart MPBART, and the Random Forest classifier, in scenarios where the underlying latent structure exhibits discontinuities.

3.2 DGP II

The data-generating process in the second setting is based on the Friedman #1 benchmark function (Friedman, 1991; Breiman, 1996), which represents a smooth, nonlinear regression function with interaction, polynomial and additive terms. We again consider a setup with three classes, for which we generate the corresponding latent functions z_1 and z_2 as variations of the Friedman function:

$$z_1 = 10 \sin(\pi x_1 x_2) - 20(x_3 - 0.5)^2 + 10x_4 + 5x_5 - 10 + \varepsilon_1, \tag{3.4}$$

$$z_2 = 8 \cos(\pi x_6 x_7) - 15(x_8 - 0.4)^2 + 7x_9 + 3x_{10} - 10 + \varepsilon_2, \tag{3.5}$$

where $x_j \overset{\text{i.i.d.}}{\sim} \text{UNIF}[0,1]$ for $j = 1, \ldots, 10$ and $\varepsilon_k \overset{\text{i.i.d.}}{\sim} N(0,1)$ for $k = 1, 2$. The class label $y \in \{0, 1, 2\}$ is determined as in Equation 3.3.

The latent functions in this setting are continuously differentiable and feature both smooth nonlinearities and interactions between predictors, making them moderate-complexity, smooth test cases. Compared to DGP I, where the latent functions contain discontinuities, this DGP provides a globally smooth and continuous alternative. This makes it valuable for evaluating S-MPBART's ability to capture nonlinear, smooth patterns and interaction effects, without the challenge of discontinuities. Note that both latent functions have been shifted downward to achieve a more balanced distribution of class labels in the training and test datasets. Table 3.2 shows that S-MPBART outperforms both benchmark methods, both in terms of accuracy and calibration. This demonstrates how S-MPBART is well-suited for classification tasks where the underlying latent functions behave in a smooth manner. Furthermore, the performance gap between S-MPBART and MPBART highlights the advantage of using soft trees in scenarios with smooth latent structures, compared to the sharp partitions induced by hard decision trees.

As a second experiment under this DGP we test how well S-MPBART is capable of distinguishing between real and noise predictors compared to the benchmark methods. In particular, we are interested in whether S-MPBART with its sparsity-inducing prior is able to select the real predictors when the predictor space is augmented with noise predictors. We do this by adding a set of 50 additional noise predictors which are generated from a standard uniform distribution. Results are shown in Table 3.2, where DGP II* denotes DGP II with the additional noise predictors included. Here we can clearly see the effect of S-MPBART's sparsity-inducing prior, as the additional noise predictors have negligible effects on its accuracy and calibration. S-MPBART is able to filter out the noise predictors well, performing effective variable selection. In contrast, both MPBART and Random Forest perform substantially worse in terms of their accuracy and confidence.

3.3 DGP III

In the final experiment, we consider a multinomial choice setting from Kindo et al. (2016) which is again based on the Friedman #1 function (Friedman, 1991). This setting considers five choice-specific predictors $\mathbf{w}_k \in \mathbb{R}^5$ for choices $k = 0, 1, 2$, which are generated from i.i.d. UNIF$[0, 1]$. Furthermore, a predictor $v \overset{\text{i.i.d.}}{\sim}$ UNIF$[0, 2]$ is included to describe the observed individual. Letting $f(\mathbf{x}) = 20\sin(\pi x_1 x_2) - 20(x_3 - 0.5)^2 + 10x_4 + 5x_5$ and $g(v) = 8v$, the latent variables are generated as

$$\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} f(\mathbf{w}_1 - \mathbf{w}_3) + g(v) \\ f(\mathbf{w}_2 - \mathbf{w}_3) + g(v) \end{pmatrix} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N\left(\mathbf{0}, \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}\right). \tag{3.6}$$

The class label again follows from Equation 3.3. Choice-specific predictors enter the function f as deviations from the reference alternative, after which the individual-specific predictor is included in an additive way. Contrary to DGP I and II, in which we assume i.i.d. error terms, in this setup we allow for moderate correlation between the error terms which creates dependence between the latent variables even after conditioning on the predictors. This more realistically captures situations where unobserved factors jointly influence multiple choices, which would be overlooked under the i.i.d. assumption. In Table 3.2 we see that S-MPBART clearly outperforms MPBART and Random Forest. The performance gap with the benchmark methods is substantial, with S-MPBART achieving a lower misclassification error, and similarly notable improvements in calibration as reflected by the Brier score. These results again show how S-MPBART leverages its soft trees to perform very well when the underlying latent structures behave smoothly.

Experiment	Method	Misclassification Error	Brier Score
DGP I	S-MPBART	0.0624 (0.0127)	0.0980 (0.0158)
	MPBART	0.0635 (0.0113)	0.0899 (0.0010)
	Random Forest	0.0673 (0.0119)	0.0999 (0.0136)
DGP II	S-MPBART	0.1425 (0.0152)	0.2049 (0.0195)
	MPBART	0.1824 (0.0155)	0.2614 (0.0186)
	Random Forest	0.2330 (0.0236)	0.3547 (0.0151)
DGP II*	S-MPBART	0.1447 (0.0101)	0.2068 (0.0157)
	MPBART	0.2276 (0.0109)	0.3337 (0.0100)
	Random Forest	0.2721 (0.0157)	0.4063 (0.0158)
DGP III	S-MPBART	0.2127 (0.0110)	0.2908 (0.0151)
	MPBART	0.3527 (0.0167)	0.4673 (0.0121)
	Random Forest	0.3602 (0.0137)	0.5035 (0.0071)

Table 3.2: Average misclassification error and Brier score (standard deviations in parentheses) across four simulation experiments, based on 10 replications for each experiment. *DGP II with an additional 50 noise predictors.

Chapter 4

Data applications

In this section we test the performance of S-MPBART on multiple real datasets. We consider both multiclass classification and multinomial choice datasets, with the key difference being that the latter contain choice-specific predictors alongside individual-specific predictors. Again Random Forest (Breiman, 2001) and MPBART as implemented in Xu et al. (2024) are used as benchmark methods.

4.1 Multiclass classification examples

The multiclass classification datasets are obtained from the UCI machine learning repository (Bache & Lichman, 2013). The datasets are summarized in Table 4.1, together with the corresponding number of training and test folds used in cross-validation.

Dataset	**# Observations**	**# Classes**	**# Predictors**	**# CV folds**
Forensic Glass	214	6	9	5
Vertebral Column	310	3	6	10
Iris	150	3	4	10
Wine	178	3	13	10
Vehicle Silhouettes	846	4	18	5
Vowel Recognition	990	11	10	5
Waveform Recognition	5000	3	21	5

Table 4.1: Summary of the multiclass classification datasets, reporting the number of observations, classes, predictors, and training and test folds used in cross-validation.

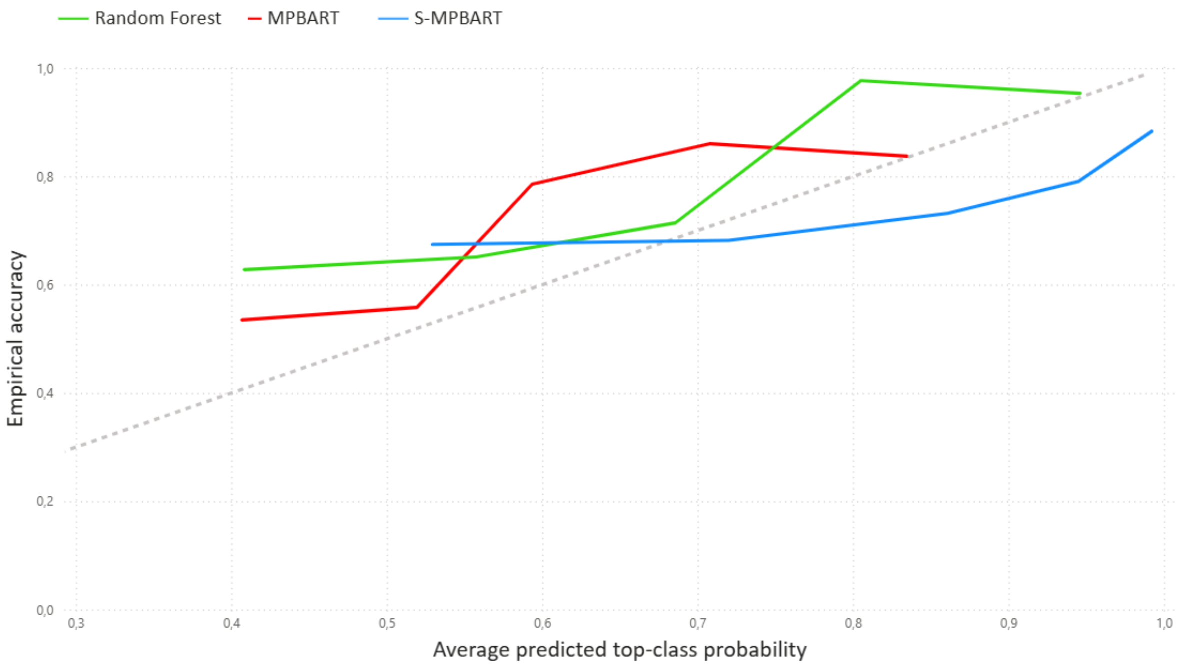
The collection of datasets varies widely in terms of size and complexity, testing the flexibility and robustness of S-MPBART across different scenarios. The number of observations ranges from 150 in the Iris dataset to 5000 in the Waveform Recognition dataset, reflecting a broad spectrum of sample sizes. The number of classes also differs substantially between datasets, from three in datasets like Iris and Wine to eleven in Vowel Recognition. While for most datasets we use ten-fold cross-validation, for those with larger sample sizes or a higher number of classes we use 5 folds to manage the computational cost of S-MPBART.

Table 4.2 shows that S-MPBART performs competitively compared to both its hard-splitting counterpart MPBART and the widely used Random Forest classifier, both in terms of accuracy and calibration. In terms of accuracy, for smaller datasets S-MPBART shows performance similar to that of MPBART, but this changes to substantial improvements when the sample size increases. Furthermore, for five out of seven datasets S-MPBART yields lower Brier scores, indicating better overall calibration of its predictions. Compared to Random Forest, we see that S-MPBART shows comparable or improved accuracy for all but the Forensic Glass and Vowel Recognition datasets, which are the two datasets with the highest number of classes. Note that the Brier scores for those datasets do not differ much.

While the Brier score provides a single measure of probabilistic performance, it combines both calibration and accuracy. As such, a low Brier score may still mask miscalibration, especially if the model is very accurate. Furthermore, the Brier score on its own does not reveal where or how a model may be miscalibrated. To complement the Brier score we show top-class calibration plots in Figure 4.1 for a selection of datasets, to gain more in-depth insight into the calibration of the models. We plot the average predicted top-class probability per bin against the corresponding empirical accuracy. Bins are constructed using quantile-based binning, which divides the predicted probabilities into intervals containing approximately equal numbers of observations. This approach provides a balanced view of calibration performance across the full range of predicted probabilities. Figure 4.1a shows the top-class calibration plot for the Forensic Glass dataset. For this dataset, Random Forest outperforms MPBART and S-MPBART in both accuracy and Brier score. However, its calibration curve reveals that it deviates more from perfect calibration than S-MPBART, illustrating how the Brier score can mask miscalibration when paired with higher predictive accuracy. Another interesting observation is that S-MPBART assigns higher predicted probabilities to its top predicted class, with the curve starting further to the right and extending all the way to an average predicted top-class probability close to one. This high level of model confidence is also visible in Figures 4.1b and 4.1c, which show calibration plots for the Vowel Recognition and Wine datasets, respectively. Across these datasets, Random Forest achieves the highest accuracy, yet its calibration curves reveal substantial underconfidence, assigning relatively low predicted probabilities to its top-class predictions given what would be expected given their actual correctness. In contrast, S-MPBART displays noticeably better calibration, with its predicted probabilities more closely aligning with observed frequencies. This demonstrates a trade-off between accuracy and probabilistic reliability, where for such cases S-MPBART may offer better-calibrated uncertainty estimates at the cost of slightly lower predictive accuracy.

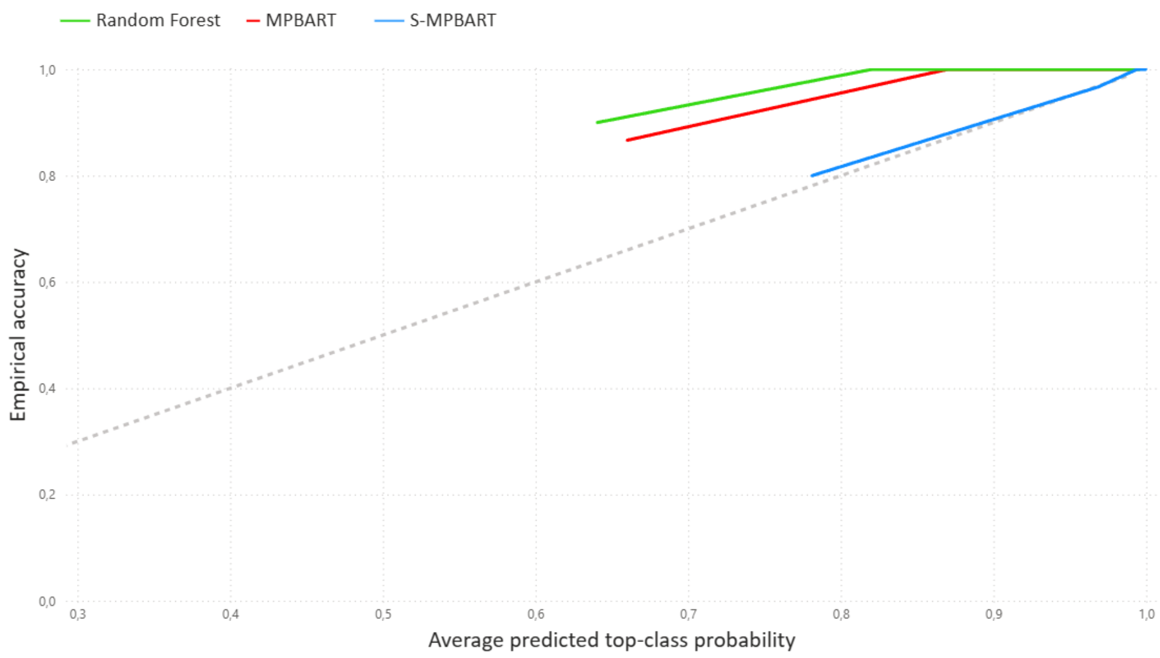
Dataset	**Method**	**Misclassification Error**	**Brier Score**
Forensic Glass	S-MPBART	0.2855 (0.0958)	0.3625 (0.1130)
	MPBART	0.2845 (0.0470)	0.4263 (0.0528)
	Random Forest	0.2055 (0.0412)	0.3403 (0.0495)
Vertebral Column	S-MPBART	0.1452 (0.0462)	0.2282 (0.0617)
	MPBART	0.1516 (0.0528)	0.2146 (0.0470)
	Random Forest	0.1419 (0.0649)	0.2145 (0.0585)
Iris	S-MPBART	0.0400 (0.0344)	0.0649 (0.0504)
	MPBART	0.0400 (0.0344)	0.0742 (0.0369)
	Random Forest	0.0533 (0.0422)	0.0729 (0.0492)
Wine	S-MPBART	0.0389 (0.0527)	0.0563 (0.0644)
	MPBART	0.0222 (0.0287)	0.0572 (0.0388)
	Random Forest	0.0167 (0.0268)	0.0637 (0.0334)
Vehicle Silhouettes	S-MPBART	0.2387 (0.0289)	0.3039 (0.0377)
	MPBART	0.2967 (0.0536)	0.3621 (0.0587)
	Random Forest	0.2482 (0.0272)	0.2985 (0.0162)
Vowel Recognition	S-MPBART	0.1343 (0.0287)	0.2011 (0.0336)
	MPBART	0.1879 (0.0254)	0.3094 (0.0134)
	Random Forest	0.0333 (0.0170)	0.1847 (0.0164)
Waveform Recognition	S-MPBART	0.1358 (0.0079)	0.2232 (0.0161)
	MPBART	0.1468 (0.0105)	0.1998 (0.0094)
	Random Forest	0.1448 (0.0065)	0.2318 (0.0046)

Table 4.2: Average misclassification error and Brier score (standard deviations in parentheses) across different multiclass classification datasets, based on the respective number of CV folds.



(a) Forensic Glass

(b) Vowel Recognition



(c) Wine

Figure 4.1: Calibration plots for the Forensic Glass (a), Vowel Recognition (b), and Wine (c) datasets. The dashed 45-degree line represents perfect calibration. Curves above the line indicate underconfidence, while curves below indicate overconfidence.

4.2 Multinomial choice examples

S-MPBART can also be applied to multinomial choice settings in which some or all predictors vary across alternatives. This can be accommodated through a straightforward modification of the predictor matrix, where all choice-specific predictors are flattened into a wide format and concatenated alongside the individual-specific predictors. To illustrate S-MPBART's predictive performance in multinomial choice settings we use two datasets that are also used in Kindo et al. (2016). The Travel Mode dataset contains for 210 individuals the choice of travel mode between Sydney and Melbourne, with options: air, train, bus or car. For each travel mode, the dataset provides information on the general cost of the option, travel time, waiting time at the terminal (not applicable for car travel), and the monetary cost of travel. It also includes as individual-specific variables the logarithm of household income and the size of the travel party. This dataset is available via the R package `AER` (Kleiber & Zeileis, 2008) and was originally discussed by Greene (2003).

The Fishing Mode dataset is based on a survey of 1182 individuals who reported their most recent saltwater fishing activity, choosing among four modes: beach, pier, boat or charter. For each option, the dataset includes as choice-specific attributes the expected catch rate per hour and the associated cost. Additionally, each individual's monthly income is included as an individual-specific variable. This dataset, originally discussed by Kling & Thomson (1996) and Herriges & Kling (1999), is available via the `mlogit` package (Croissant, 2020) in R. For the Travel and Fishing Mode datasets, we use ten-fold and five-fold cross-validation, respectively.

Table 4.3 shows that S-MPBART displays competitive performance in terms of accuracy on both datasets. Especially for the Travel Mode dataset, S-MPBART substantially outperforms the benchmark methods, suggesting that leveraging smoothness is crucial for achieving strong performance on this dataset. These results further demonstrate S-MPBART's potential as a flexible, general-purpose classifier.

Dataset	**Method**	**Misclassification Error**	**Brier Score**
Travel Mode	S-MPBART	0.0238 (0.0405)	0.0329 (0.0372)
	MPBART	0.1056 (0.0562)	0.2119 (0.0569)
	Random Forest	0.0526 (0.0529)	0.0941 (0.0532)
Fishing Mode	S-MPBART	0.4619 (0.0242)	0.6431 (0.0452)
	MPBART	0.4611 (0.0411)	0.5389 (0.0227)
	Random Forest	0.5009 (0.0309)	0.6123 (0.0370)

Table 4.3: Average misclassification error and Brier score (standard deviations in parentheses) across different multinomial choice datasets, based on the respective number of CV folds.

Chapter 5

Discussion

We have introduced soft multinomial probit BART (S-MPBART) as a novel extension of BART for multiclass classification. By integrating soft decision trees in the multinomial probit BART framework, S-MPBART allows both individual- and choice-specific predictors to have smooth effects on the class probabilities. Through multiple simulation studies, we find that S-MPBART shows strong predictive performance compared to existing methods when the underlying latent structures behave smoothly, while remaining competitive even in the absence of such smoothness. Moreover, S-MPBART achieves effective variable selection through its use of a sparsity-inducing prior. In real data examples, S-MPBART's predictive performance is competitive with existing methods, while consistently outperforming MPBART.

A distinguishing feature of S-MPBART is its ability to naturally accommodate both individual-specific and choice-specific predictors in a utility-based framework, which is typically not supported by standard classifiers. In addition, by using a sparsity-inducing prior on splitting proportions, S-MPBART enables implicit variable selection within a fully Bayesian framework, allowing it to automatically identify the most relevant predictors. Together, these capabilities allow S-MPBART to capture complex predictor structures while remaining interpretable and widely applicable.

Among its limitations, S-MPBART requires a notably higher computational cost compared to other classifiers. This is primarily because sampling soft decision trees is more computationally intensive than the already demanding sampling of hard-split trees in MPBART. In addition to the longer runtime, we have yet to investigate how S-MPBART handles categorical predictors. Future work could explore alternative sampling strategies to improve computational efficiency—such as those proposed by Xu et al. (2024) for MPBART—and incorporate flexible approaches for categorical predictors like Deshpande (2024) within the S-MPBART framework.

References

Albert, J. H. & Chib, S. (1993). Bayesian analysis of binary and polychotomous response data. *Journal of the American Statistical Association*, *88*(422), 669–679. doi: 10.1080/01621459.1993.10476321

Athreya, K. B. & Ney, P. E. (2004). *Branching processes.* New York: Dover Publications.

Bache, K. & Lichman, M. (2013). *Uci machine learning repository.* `http://archive.ics.uci.edu/ml`. (University of California, Irvine, School of Information and Computer Sciences)

Bonato, V., Baladandayuthapani, V., Broom, B. M., Sulman, E. P., Aldape, K. D. & Do, K.-A. (2011). Bayesian ensemble methods for survival prediction in gene expression data. *Bioinformatics*, *27*(3), 359–367.

Bouazizi, M. & Ohtsuki, T. (2019). Multi-class sentiment analysis on twitter: Classification performance and challenges. *Big Data Mining and Analytics*, *2*(3), 181–194.

Breiman, L. (1996). Bagging predictors. *Machine Learning*, *24*(2), 123–140. Retrieved from `https://doi.org/10.1007/BF00058655` doi: 10.1007/BF00058655

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. doi: 10.1023/A:1010933404324

Breiman, L., Friedman, J. H., Olshen, R. A. & Stone, C. J. (1984). *Classification and regression trees.*

Bunker, R. P. & Thabtah, F. (2019). A machine learning framework for sport result prediction. *Applied Computing and Informatics*, *15*(1), 27-33. Retrieved from `https://www.sciencedirect.com/science/article/pii/S2210832717301485` doi: https://doi.org/10.1016/j.aci.2017.09.005

Burgette, L. F. & Nordheim, E. V. (2012). The trace restriction: An alternative identification strategy for the bayesian multinomial probit model. *Journal of Business & Economic Statistics*, *30*(3), 404–410.

Chakraborty, S. (2016). Chapter 7 - bayesian additive regression tree for seemingly unrelated regression with automatic tree selection. In V. N. Gudivada, V. V. Raghavan, V. Govindaraju & C. Rao (Eds.), *Cognitive computing: Theory and applications* (Vol. 35, p. 229-251). Elsevier. Retrieved from `https://www.sciencedirect.com/science/article/pii/S0169716116300487` doi: https://doi.org/10.1016/bs.host.2016.07.007

Chipman, H. A., George, E. I. & McCulloch, R. E. (1998). Bayesian cart model search. *Journal of the American Statistical Association*, *93*, 935–960.

Chipman, H. A., George, E. I. & McCulloch, R. E. (2010). Bart: Bayesian additive regression trees. *The Annals of Applied Statistics*, *4*(1), 266–298.

Croissant, Y. (2020). Estimation of random utility models in R: The mlogit package. *Journal of Statistical Software*, *95*(11), 1–41. doi: 10.18637/jss.v095.i11

Deshpande, S. K. (2024). *flexbart: Flexible bayesian regression trees with categorical predictors.* Retrieved from `https://arxiv.org/abs/2211.04459`

Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J. J., Sandhu, S., ... Froelicher, V. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. *The American Journal of Cardiology*, *64*(5), 304–310. doi: 10.1016/0002-9149(89)90524-9

Fisher, R. A. (1936). The use of multiple measurements in taxonomic problems. *Annals of eugenics*, *7*(2), 179–188.

Friedman, J. H. (1991). Multivariate adaptive regression splines. *The Annals of Statistics*, *19*(1), 1–67. doi: 10.1214/aos/1176347963

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, *29*(5), 1189–1232. doi: 10.1214/aos/1013203451

Greene, W. H. (2003). *Econometric analysis.* Pearson education india.

Hastie, T. J. & Tibshirani, R. (2000). Bayesian backfitting (with comments and a rejoinder by the authors). *Statistical Science*, *15*(3), 196–223. doi: 10.1214/ss/1009213723

Herriges, J. A. & Kling, C. L. (1999). Nonlinear income effects in random utility models. *Review of Economics and Statistics*, *81*(1), 62–72.

Huber, F. & Rossini, L. (2022). Inference in bayesian additive vector autoregressive tree models. *The Annals of Applied Statistics*, *16*(1), 104–123.

Imai, K. & van Dyk, D. A. (2005). A bayesian analysis of the multinomial probit model using marginal data augmentation. *Journal of Econometrics*, *124*(2), 311–334.

Irsoy, O., Yildiz, O. T. & Alpaydin, E. (2012). Soft decision trees. In *Proceedings of the 21st international conference on pattern recognition (icpr2012)* (pp. 1819–1822).

Kansal, T., Bahuguna, S., Singh, V. & Choudhury, T. (2018). Customer segmentation using k-means clustering. In *2018 international conference on computational techniques, electronics and mechanical systems (ctems)* (p. 135-139). doi: 10.1109/CTEMS.2018.8769171

Kapelner, A. & Bleich, J. (2016). bartmachine: Machine learning with bayesian additive regression trees. *Journal of Statistical Software*, *70*(4), 1–40. doi: 10.18637/jss.v070.i04

Keane, M. P. (1992). A note on identification in the multinomial probit model. *Journal of Business & Economic Statistics*, *10*(2), 193–200.

Kindo, B. P., Wang, H. & Peña, E. A. (2016). Multinomial probit bayesian additive regression trees. *Stat*, n/a–n/a. Retrieved from http://dx.doi.org/10.1002/sta4.110 doi: 10.1002/sta4.110

Kleiber, C. & Zeileis, A. (2008). *Applied econometrics with r*. Springer Science & Business Media.

Kling, C. L. & Thomson, C. J. (1996). The implications of model specification for welfare estimation in nested logit models. *American Journal of Agricultural Economics*, *78*(1), 103–114.

Liaw, A. & Wiener, M. (2002). Classification and regression by randomforest. *R News*, *2*(3), 18-22. Retrieved from https://CRAN.R-project.org/doc/Rnews/

Linero, A. R. (2018). Bayesian regression trees for high-dimensional prediction and variable selection. *Journal of the American Statistical Association*, *113*(522), 626–636. doi: 10.1080/01621459.2016.1260469

Linero, A. R. (2022). Softbart: Soft bayesian additive regression trees. *arXiv e-prints*.

Linero, A. R. & Yang, Y. (2018). Bayesian regression tree ensembles that adapt to smoothness and sparsity. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, *80*(5), 1087–1110. Retrieved from https://doi.org/10.1111/rssb.12293 doi: 10.1111/rssb.12293

McCulloch, R. E., Polson, N. G. & Rossi, P. E. (2000). A bayesian analysis of the multinomial probit model with fully identified parameters. *Journal of Econometrics*, *99*(1), 173–193.

McCulloch, R. E. & Rossi, P. E. (1994). An exact likelihood analysis of the multinomial probit model. *Journal of Econometrics*, *64*(1), 207–240.

McFadden, D. (1973). Conditional logit analysis of qualitative choice behavior. , 105–142.

McFadden, D. (1989). A method of simulated moments for estimation of discrete response models without numerical integration. *Econometrica: Journal of the Econometric Society*, 995–1026.

O'Neill, E. (2024). Type i tobit bayesian additive regression trees for censored outcome regression. *Statistics and Computing*, *34*(4), 123.

Pak, M. & Kim, S. (2017). A review of deep learning in image recognition. In *2017 4th international conference on computer applications and information processing technology (caipt)* (p. 1-3). doi: 10.1109/CAIPT.2017.8320684

Park, C. (2022). Jump gaussian process model for estimating piecewise continuous regression functions. *Journal of Machine Learning Research*, *23*(278), 1–37.

Quinlan, J. R. (1986). Induction of decision trees. *Machine learning*, *1*, 81–106.

Sajda, P. (2006). Machine learning for detection and diagnosis of disease. *Annu. Rev. Biomed. Eng.*, *8*(1), 537–565.

Shen, D., Wu, G. & Suk, H.-I. (2017). Deep learning in medical image analysis. *Annual review of biomedical engineering*, *19*(1), 221–248.

Tan, Y. & Roy, J. (2019). Bayesian additive regression trees and the general bart model. *Statistics in Medicine*, *38*(25), 5048–5069. doi: 10.1002/sim.8344

van Wezel, M. & Potharst, R. (2007). Improved customer choice predictions using ensemble methods. *European Journal of Operational Research*, *181*(1), 436-452. Retrieved from `https://www.sciencedirect.com/science/article/pii/S0377221706003900` doi: https://doi.org/10.1016/j.ejor.2006.05.029

Vehtari, A., Gelman, A. & Gabry, J. (2017). Practical bayesian model evaluation using leave-one-out cross-validation and waic. *Statistics and Computing*, *27*(5), 1413–1432.

Watanabe, S. (2013). A widely applicable bayesian information criterion. *Journal of Machine Learning Research*, *14*, 867–897.

Xu, Y., Hogan, J., Daniels, M., Kantor, R. & and, A. M. (2024). Augmentation samplers for multinomial probit bayesian additive regression trees. *Journal of Computational and Graphical Statistics*, *0*(0), 1–11. Retrieved from `https://doi.org/10.1080/10618600.2024.2388605` doi: 10.1080/10618600.2024.2388605

Yu, D. & Deng, L. (2016). *Automatic speech recognition* (Vol. 1). Springer.

Zhang, J. L. & Härdle, W. K. (2010). The bayesian additive classification tree applied to credit risk modelling. *Computational Statistics & Data Analysis*, *54*(5), 1197–1205.

Appendix A

Run settings

The run settings for S-MPBART are the same for all simulation experiments and data applications, and are provided in Table A.1. Any other parameters which are not mentioned have the default specification from the `SoftBART` package (Linero, 2022) in R.

Initialization Parameters	
Tree structures ($\boldsymbol{\mathcal{T}}$)	Single-node trees
Leaf node parameters ($\boldsymbol{\mathcal{M}}$)	Initialized at 0
Splitting probabilities ($\mathbf{s}_k$)	Uniform over all predictors for all k
Bandwidths (τ_{kj})	Initialized at 0.1 for all k, j
$\boldsymbol{\Sigma}$	I_K
Fixed Parameters	
MCMC iterations (n_sim)	1500
Burn-in iterations (n_burnin)	1500
Number of trees (m)	200
Gating function (ψ)	$\psi(x) = 1/(1 + e^{-x})$

Table A.1: Overview of S-MPBART initialization and hyperparameter settings.

For MPBART we use the `GcompBART` package in R, which implements MPBART as in Xu et al. (2024). We use the default settings of this package, along with 1500 MCMC iterations, 1500 burn-in iterations and $m = 200$ trees, across all simulations and real datasets.

Finally, for the implementation of the Random Forest classifier we use the `randomForest` package (Liaw & Wiener, 2002) in R with default settings. We use cross-validation to tune the `mtry` parameter, and provide the grids per simulation setting and dataset in Table A.2.

Setting	`mtry` **grid**
Simulations	
DGP I	{1, 2}
DGP II	{2, 3, 4, 6, 8, 10}
DGP II*	{2, 4, 8, 15, 30, 60}
DGP III	{3, 4, 5, 8, 12, 16}
Data Applications	
Forensic Glass	{2, 3, 4, 6, 7, 9}
Vertebral Column	{2, 3, 4, 6}
Iris	{1, 2, 3, 4}
Wine	{2, 3, 4, 5, 8, 12}
Vehicle Silhouettes	{2, 4, 5, 6, 10, 14, 18}
Vowel Recognition	{2, 3, 4, 6, 10}
Waveform Recognition	{2, 4, 5, 6, 10, 15, 21}
Travel Mode	{2, 3, 4, 8, 12, 18}
Fishing Mode	{2, 3, 4, 6, 9}

Table A.2: Grid of `mtry` values used for Random Forest in simulation experiments and data applications.